

IIT BOMBAY
CHEMICAL ENGINEERING

CL420: Tutorial Problems

Prof. Pramod P Wangikar

Practice Questions

Q1 — Black box & yield coefficients

A batch culture consumes 12 g of glucose and produces 4 g of new cell mass. Calculate the observed growth yield coefficient $Y_{x/s}$ (g cells / g substrate).

Q2 — Substrate partitioning (Eqs. 5.47–5.48)

Of the 12 g glucose consumed above, 1 g is later found to have gone to maintenance energy alone. Calculate the “true” (growth-associated) yield coefficient $Y_{x/s,true}$.

Q3 — Elemental balance (5.10.2)

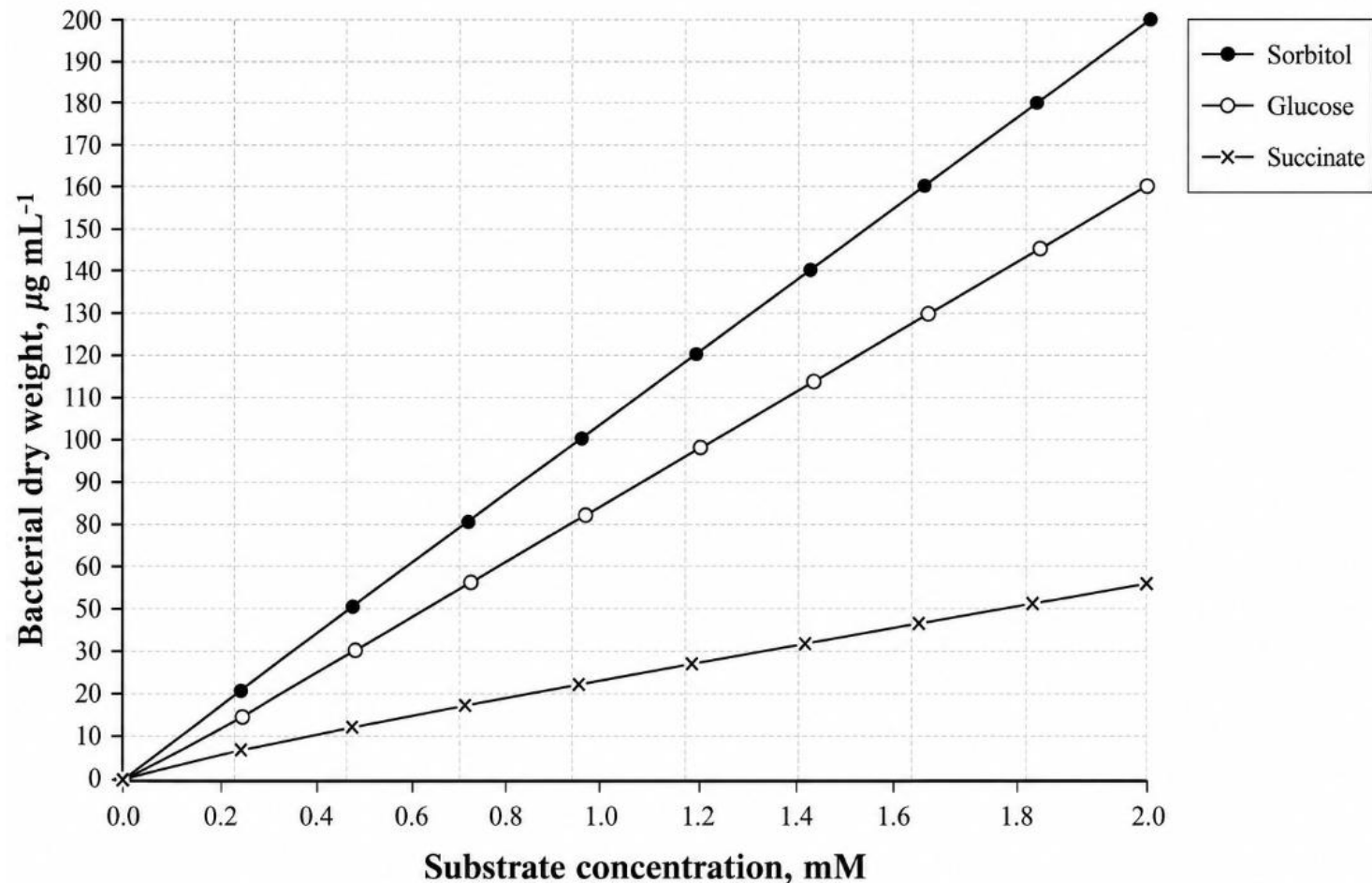
Per C-mole of glucose (CH_2O) consumed in aerobic growth, 0.5 C-mole of biomass ($\text{CH}_{1.8}\text{O}_{0.5}\text{N}_{0.2}$) is formed with no separate product. Using a carbon balance alone, find the moles of CO_2 released per C-mole glucose.

Q4 — Product formation (5.10.3)

Anaerobic yeast fermentation converts glucose (CH_2O per C) to biomass ($\text{CH}_{1.8}\text{O}_{0.5}\text{N}_{0.2}$, 0.10 C-mol/C-mol glucose), ethanol ($\text{CH}_3\text{O}_{0.5}$ per C, 0.85 C-mol/C-mol glucose), and CO_2 . Find the CO_2 yield using a carbon balance.

Q5 — Calculating $Y_{x/s}$ from given data

From the aerobic *Aerobacter aerogenes* glucose line in Fig. 5.29 (0 to 2 mM $\rightarrow$ 0 to 150 $\mu\text{g mL}^{-1}$), estimate $Y_{x/s}$ in $\mu\text{g mL}^{-1}$ per mM and in g/g. MW of sorbitol = 182 and succinic acid is 118.



Q6 — Single-Cell Protein from Ethane

Nielsen, Villadsen & Lidén, Problem 3.2 — empirical formula, O_2 demand, and identifying an unmeasured product.

Mycobacterium vaccae grows on ethane (C_2H_6) as its sole carbon and energy source, with NH_3 as the nitrogen source. Ethane is the limiting substrate, and the yield $Y_{sx} = 23.7$ g dry weight per mole ethane.

- (a) Elemental analysis of dry cell mass (excluding small amounts of S, P): C 47.4 wt%, N 8.30 wt%, H 7.43 wt%, ash 4.00 wt% — O is found by difference. Determine the elemental composition of the ash-free biomass and its formula weight per C-atom. Calculate Y_{sx} in C-mol $(C\text{-mol})^{-1}$.
- (b) Calculate the oxygen consumption Y_{so} (mol O_2 per C-mol ethane) assuming CO_2 is the only metabolic product besides biomass and H_2O . Write the stoichiometry for conversion of ethane to SCP.
- (c) Measurements show 10.44 g DW formed per mole O_2 consumed — the same Y_{sx} as above — but $RQ = 0.3805$. Show that an extra metabolic product must be formed besides CO_2 (and H_2O). Determine its degree of reduction, and suggest its identity.

Q7 — Checking an Anaerobic Mass Balance

Nielsen, Villadsen & Lidén, Problem 3.4 — the same consistency-check idea as Problem 5.9, on real chemostat data.

In a steady-state, strictly anaerobic chemostat ($V = 1 \text{ L}$), *S. cerevisiae* grows on a sterile medium with 23 g L^{-1} glucose and $1 \text{ g L}^{-1} \text{ NH}_3$, fed at $v = 0.1 \text{ L h}^{-1}$. Biomass composition: $X = \text{CH}_{1.76}\text{O}_{0.56}\text{N}_{0.17}$.

Effluent (g L^{-1}): glucose 0.048 ; NH_3 0.728 ; biomass 2.36 ; ethanol 8.85 ; glycerol 2.39 . Exit gas $\text{CO}_2 = 2.88 \text{ vol\%}$, total gas flow 0.25 L min^{-1} (30°C , 1 atm ; feed gas flow assumed $\approx$ exit gas flow).

- (a) Use the data to set up a carbon balance. Show that the data are not consistent, and calculate the yield coefficient of the missing metabolite.

- (b) Calculate the degree of reduction of the missing metabolite. What do you think is the source of the error?

Q8 — The Case of the Missing Carbon (Respiro-Fermentative Yeast)

Nielsen, Villadsen & Lidén, Example 3.5 continued — same chemostat as before, above the critical dilution rate.

Above $D_{crit} \approx 0.25 \text{ h}^{-1}$ the same aerobic *S. cerevisiae* chemostat (von Meyenburg, 1969) shifts to respiro-fermentative growth and starts producing ethanol. Biomass: $X = \text{CH}_{1.83}\text{O}_{0.56}\text{N}_{0.17}$ ($\gamma_x = 4.20$). Measured yield coefficients (C-mol/C-mol glucose):

D (h^{-1})	Y_{sx}	Y_{sp} (ethanol)	Y_{sc} (CO_2)	Y_{so} (O_2)
0.3	0.279	0.196	0.366	0.167
0.4	0.175	0.394	0.312	0.044

- (a) Show that carbon is missing from both balances, and calculate the yield coefficient Y_{sp_1} of the missing product at each D.
- (b) Use the degree-of-reduction balance to find the degree of reduction k_{p_1} of the missing product at each D. What is its likely identity — and why might it have gone undetected in the liquid-phase assay?

Q9 — Citric Acid and a Toxic Contaminant

Nielsen, Villadsen & Lidén, Example 3.6 — a new N-source, and a second missing-product hunt.

Citric acid ($\text{C}_6\text{H}_8\text{O}_7$) is produced by fermenting glucose with *Aspergillus niger*, using NH_4NO_3 as the nitrogen source. Biomass: $\text{X} = \text{CH}_{1.8}\text{O}_{0.5}\text{N}_{0.2}$ ($\gamma_x = 4.20$). Nitrate contributes $\kappa\text{N} = -8$ to the electron balance (it's more oxidized than the NH_3 reference).

(a) At pH 1.8–2 (normal operation), 68 g citric acid forms per 100 g glucose consumed, and practically no CO_2 is produced. Given Y_{sp} (citric acid) = 0.6375 C-mol/C-mol glucose, use a carbon balance to find Y_{sx} , then the degree-of-reduction balance to find Y_{so} .

(b) At pH 4–5, citric acid yield drops to 0.5375 C-mol/C-mol (Y_{sx} unchanged at 0.3625) but Y_{so} rises to 0.11875 mol O_2 /C-mol glucose. Show a second product must form, find its yield and degree of reduction, and suggest its identity — and why producers keep pH below 4.

Solutions — Q1 & Q2

Both come straight from the yield-coefficient and substrate-partitioning definitions.

Q1 — Observed yield coefficient

$$Y_{x/s} = \Delta X / \Delta S = 4 \text{ g} / 12 \text{ g}$$

$$Y_{x/s} \approx 3.33 \times 10^{-1} \text{ g/g}$$

As expected, $Y_{x/s, \text{true}} > Y_{x/s, \text{obs}}$. i.e., stripping out the maintenance-only substrate makes the culture look more efficient at growth.

Q2 — True (growth-only) yield coefficient

By Eq. 5.47, only 11 g of the 12 g consumed went to assimilation and growth energy (the remaining 1 g was maintenance); that 11 g still produced all 4 g of cells.

$$Y_{x/s, \text{true}} = \Delta X / (\Delta S - \Delta S_{\text{maint}}) = 4 / (12 - 1)$$

$$Y_{x/s, \text{true}} = 4/11 \approx 3.64 \times 10^{-1} \text{ g/g}$$

Solution — Q3

Set up (per 1 C-mol glucose):



Carbon balance (asked):

$$1 = 0.5(1) + e' \rightarrow e' = 0.5$$

CO₂ released, e' = 0.5 mol per C-mol glucose

Rigor check — N, H, O balances (not required by Q3, shown for completeness):

$$\text{N: } c' = 0.5(0.2) = 0.10$$

$$\text{H: } 2 + 3(0.10) = 0.5(1.8) + 2d' \rightarrow d' = 0.70$$

$$\text{O: } 1 + 2b' = 0.5(0.5) + d' + 2e' \rightarrow b' = 0.475$$

Electron-balance cross-check:

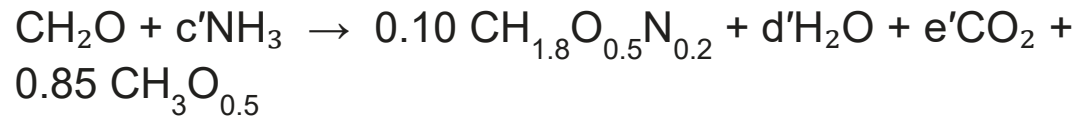
$$Y_{\text{glucose}} = 4.0: 1(4) = 0.5(4.2) + 4(0.475) = 2.1 + 1.9 = 4.0 \quad \checkmark$$

Same method as Problem 2.7: the elemental balances alone can leave one coefficient (here, O₂) undetermined by the carbon and nitrogen data; the electron balance closes it exactly, matching the H and O balances and independently confirming that the whole system is self-consistent.

Solution — Q4

Same carbon-balance mechanics as Q3, now with a product term (5.10.3).

Set up (per 1 C-mol glucose, anaerobic — no O₂):



Carbon balance (asked):

$$1 = 0.10(1) + e' + 0.85(1) \rightarrow e' = 1 - 0.10 - 0.85$$

CO₂ released, e' = 5×10⁻² mol per C-mol glucose

As in 5.10.2, one balance (here carbon) is enough to isolate the one unknown asked for; the full system (C, H, O, N + electrons) is what you'd need to pin down every coefficient at once, that's the Problem 5.9 approach.

Caveat: these yields (0.10, 0.85) were chosen to be round for a clean carbon-balance answer they aren't necessarily electron-balance-consistent (that would also fix d' and constrain β and φ together). Problem 5.9 works a fully self-consistent version of exactly this calculation, cross-checked with the electron balance.

Solution — Q5

Reading a yield coefficient straight off Fig. 5.29, the slope is the answer.

Slope between the two endpoints of the glucose line:

$$Y_{X/S} = \Delta(\text{dry wt}) / \Delta S = (150 - 0) \mu\text{g mL}^{-1} / (2 - 0) \text{ mM}$$

$$Y_{X/S} \approx 7.5 \times 10^1 \mu\text{g mL}^{-1} \text{ mM}^{-1}$$

Sorbitol gives roughly 3× the yield of succinate for the same substrate concentration, the same pattern, the degree-of-reduction discussion predicts: more-reduced substrates (higher γ) support more biomass per mole.

For comparison, the same reading on the other two lines:

$$\text{Sorbitol: } (195-0)/(2-0) \approx 9.8 \times 10^1 \mu\text{g mL}^{-1} \text{ mM}^{-1}$$

$$\text{Succinate: } (65-0)/(2-0) \approx 3.3 \times 10^1 \mu\text{g mL}^{-1} \text{ mM}^{-1}$$

Solution — Q6 (a)-(b)

Same elemental-analysis method as Problem 2.8(a); same growth-balance method as 5.10.2.

(a) Empirical formula from elemental analysis (basis: 100 g dry biomass):

$$\text{O} = 100 - 47.4 - 7.43 - 8.30 - 4.00 \text{ (ash)} = 32.87 \text{ g} \rightarrow \text{mol C:H:N:O} = 3.946 : 7.371 : 0.593 : 2.054$$

$$\text{Normalized to C} = 1: \text{X} = \text{CH}_{1.87}\text{O}_{0.52}\text{N}_{0.15} \quad (\text{FW} \approx 24.3 \text{ g/C-mol})$$

$$\text{Y}_{\text{sx}} = (23.7 \text{ g} / 2 \text{ C-mol ethane}) / 24.3 \text{ g/C-mol} \approx 4.87 \times 10^{-1} \text{ C-mol/C-mol}$$

(b) Growth-only balance (ethane per C-mol = CH_3 ; yethane = $4(1)+3 = 7$):

$$\text{Carbon: } \text{Y}_{\text{sc}} = 1 - \text{Y}_{\text{sx}} = 1 - 0.487 = 0.513$$

$$\text{Electrons: } 7(1) = \gamma_{\text{x}} \cdot \text{Y}_{\text{sx}} + 4 \cdot \text{Y}_{\text{so}} \rightarrow 7 = 4.376(0.487) + 4\text{Y}_{\text{so}}$$

$$\text{Y}_{\text{so}} \approx 1.22 \text{ mol O}_2 / \text{C-mol ethane}$$



(Y_{sn}, Y_{sw} from the N and O balances — not asked, shown for a complete stoichiometry)

Solution — Q6 (c)

RQ disagrees with the “CO₂-only” case in (b) — the gap has to be a real product.

Convert the measured O₂-basis yield to a substrate-basis Y_{so}:

$$Y_{xo} = 10.44 / 24.3 = 0.429 \text{ C-mol/mol O}_2 \rightarrow Y_{so} = Y_{sx} / Y_{xo} = 0.487 / 0.429 \approx 1.135 \text{ mol O}_2/\text{C-mol}$$

$$RQ = Y_{sc} / Y_{so} = 0.3805 \rightarrow Y_{sc} = 0.3805 \times 1.135 = 0.432 \text{ C-mol/C-mol}$$

But part (b) needed Y_{sc} = 0.513 for CO₂ to be the only other product — 0.432 is short. By carbon balance, the missing product P must carry:

$$Y_{ps} = 1 - 0.487 - 0.432 = 8.1 \times 10^{-2} \text{ C-mol/C-mol}$$

Electron balance solves for the unknown γ of P:

$$7 = 4.376(0.487) + \gamma_p(0.081) + 4(1.135) \rightarrow \gamma_p \approx 4.05$$

$\gamma_p \approx 4.05$ is essentially exactly acetic acid's degree of reduction ($\gamma = 4/\text{C}$, Table 3.3) — and acetate overflow is a well-documented feature of hydrocarbon-oxidizing bacteria like *M. vaccae*, so acetic acid is the most chemically sensible identity for the unmeasured product.

Solution — Q7

Convert every stream to C-mol/h, then run the same carbon + electron checks as Problem 5.9.

Rates (C-mol/h), from $v=0.1$ L/h and gas flow 15 L/h at 30°C:

Glucose consumed: 7.65×10^{-2}

Biomass: 9.40×10^{-3}

Ethanol: 3.84×10^{-2}

Glycerol: 7.79×10^{-3}

CO₂ (from 2.88% $\times$ 0.603 mol/h gas): 1.74×10^{-2}

Carbon out = 0.954 of glucose in — 4.6% short.

$Y_{\text{missing}} \approx 4.63 \times 10^{-2}$ C-mol/C-mol glucose

(b) Electron balance ($\gamma_{\text{glc}}=4.0$, $\gamma_{\text{x}}=4.13$, $\gamma_{\text{e}}=6.0$, $\gamma_{\text{g}}=4.67$):

$$4.0 \approx 4.13(0.123) + 6.0(0.502) + 4.67(0.102) + \gamma_{\text{missing}}(0.0463)$$

$\gamma_{\text{missing}} \approx 0.10$ — nearly indistinguishable from 0

A real metabolite this oxidized would be unusual — organic products typically run $\gamma = 3$ to 7 (Table 3.3). $\gamma \approx 0$ points instead to a small systematic measurement error rather than a genuine unmeasured product.

The likeliest culprit: the exit gas flow rate. It's the one measurement here taken from a single flow-meter reading with no cross-check — exactly the “instrument drift” failure mode already flagged for low gas rates.

Solution — Q8

Same method as Q6(c) — carbon balance finds the gap, the electron balance identifies it.

(a) Carbon balance: $Y_{sp_1} = 1 - Y_{sx} - Y_{sp} - Y_{sc}$

$$D=0.3: Y_{sp_1} = 1 - 0.279 - 0.196 - 0.366 = 0.159$$

$$D=0.4: Y_{sp_1} = 1 - 0.175 - 0.394 - 0.312 = 0.119$$

(b) Electron balance: $4 = \gamma_x \cdot Y_{sx} + 6 \cdot Y_{sp} + \kappa_{p_1} \cdot Y_{sp_1} + 4 \cdot Y_{so}$

$$D=0.3: 4 = 4.20(0.279) + 6(0.196) + \kappa_{p_1}(0.159) + 4(0.167)$$

$$D=0.4: 4 = 4.20(0.175) + 6(0.394) + \kappa_{p_1}(0.119) + 4(0.044)$$

$$\kappa_{p_1} \approx 6.19 \text{ (D=0.3) and } 6.09 \text{ (D=0.4)}$$

Sanity check: the loss shrinks from 16% of glucose-C at $D=0.3$ to 12% at $D=0.4$ — consistent with stripping, since the rate of stripping is roughly independent of D while ethanol production rate rises with D , so a smaller fraction is lost at higher D .

This is the same pattern already flagged in “When Stoichiometric Data Goes Wrong” — here proven with real numbers instead of just named as a risk.

Both ≈ 6 — ethanol's γ (Table 3.3). P_1 is additional ethanol never captured by the liquid-phase assay: even a small liquid concentration of a volatile compound like ethanol corresponds to a large gas-phase loss when N_2 /air is sparged through — exactly the “gas stripping” failure mode.

Solution — Q9

Same detective method again — this time the culprit is a well-known citric-acid contaminant.

(a) Carbon balance (no CO₂): $Y_{sx} = 1 - Y_{sp} = 1 - 0.6375 = 0.3625$

N balance: $Y_{sn} = 0.1 \cdot Y_{sx} = 0.03625$ (citric acid $\gamma_p = 3.0$)

Electrons: $-4 + 4Y_{so} - Y_{sn} \cdot \kappa_N + Y_{sx} \cdot \gamma_x + Y_{sp} \cdot 3 = 0$

$-4 + 4Y_{so} - (0.03625)(-8) + 0.3625(4.20) + 0.6375(3) = 0$

$Y_{so} \approx 6.88 \times 10^{-2}$ mol O₂ / C-mol glucose

(b) Carbon balance: $Y_{sp_1} = 1 - 0.3625 - 0.5375 = 0.1$

Electrons: $-4 + 4(0.11875) - (0.03625)(-8) + 0.3625(4.20) + 0.5375(3) + 0.1\kappa_{p_1} = 0$

$\kappa_{p_1} = 1.0$

$\gamma = 1.0$ matches oxalic acid (Table 3.3) exactly. It forms via a TCA side-reaction (pyruvate → oxaloacetate → oxalate) favored above pH 4 — and it's toxic, so producers hold pH below 4 to suppress it.

New wrinkle vs. Q6/Q8: the nitrogen source itself (NO₃⁻) now carries electrons too — the balance needs a κ_N term the earlier problems didn't.

A real gene-deletion strain (removing the enzyme that makes oxalate) later solved this in industry, without hurting citric acid yield or growth rate — turning a stoichiometric diagnosis into a strain-engineering fix.

Key Takeaways

The black box simplifies everything

Treat the cell as a fixed catalyst: substrates in, more cells + products out. Every other idea in this deck refines that one picture.

Yield coefficients are what you actually measure

$Y_{x/s}$, $Y_{p/s}$, Y_{x/O_2} , RQ — simple ratios, and Fig. 5.29 shows they're just slopes of real growth data.

Not all substrate builds cells

Eqs. 5.47–5.48 split consumption into assimilation, growth energy, and maintenance — which is why $Y_{x/s,obs} \neq Y_{x/s,true}$.

Elemental + electron balances make it rigorous

4 atom balances (C,H,O,N) plus 1 electron (γ) balance turn yield ratios into a fully solvable stoichiometric system :Sec. 5.10.2–5.10.3.

Missing data always comes from one measurement

Whether it's Problem 2.7's respiratory quotient, Q4's product yield, or Problem 5.9's ethanol. One experimental number closes the last degree of freedom.